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A tool for rapid assessment of product usability and universal design: Development and preliminary psychometric testing

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Abstract. *Background*: While there are many available tools and methods to evaluate product usability, few have been tested on user groups with disabilities and even fewer systematically consider universal design principles.
Objective: This paper describes the development and preliminary psychometric testing of the Rapid Assessment of Product Usability & Universal Design (RAPUUD), a 12-item user-report tool based on the seven principles of universal design.
Method: A preliminary set of items was created to elicit ratings of diverse product characteristics (e.g., physical effort, cognitive effort, assistance required, safety). Data were gathered from 61 participants who rated the usability of products they use in their own environments.
Results: Each item elicited a full range of responses, with no apparent floor or ceiling effects. Collectively, the 12 items achieved a high internal consistency (Cronbach's $\alpha = 0.80$). The data indicate that the tool was sensitive to differences in functional abilities, as well as differences in product characteristics. The instrument was usable for a range of consumer products, though not all items were appropriate for each and every product.
Conclusion: The results suggest that the instrument could become a pragmatic tool for designers to identify usability problems experienced by a diversity of user populations.

Keywords: Universal design, usability, consumer products, measurement, special populations, disability

1. Introduction

The prevalence of disability is expected to increase as the U.S. population ages. Although nearly 75% of those over age 75 have a functional limitation affecting their ability to use products [12], product designers do not routinely consider the impacts of disability and aging on the usability of their products [18,23]. Older adults and wheelchair users report numerous usability challenges with products in their kitchens, bathrooms, and home offices [15,16]. Thus, there is a need to improve the usability of the products that are being created and sold in the marketplace.

Universal design (UD) is one approach that has been proposed to address this need. It has been defined as, "... an approach to design that incorporates products as well as building features which, to the greatest extent feasible, can be used by everyone" [17]. Its seven prin-

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ciples [5] and overarching philosophy have been enthusiastically received by academics and advocates in the rehabilitation technology and environmental design communities. Accessibility considerations are relevant at each step in the product development process, including conceptualization, design, evaluation, and manufacture [9]. Universal design is intended to improve the lives of people with disabilities by providing guidelines for product developers [4].

Unfortunately, current UD resources are insufficient to meet the needs of product designers and developers. Law et al. [14] reviewed eight UD and accessibility resources and identified five common problems: (a) UD resources do not consistently define their target audience or its needs; (b) the terminology used for accessible and universal design is imprecise; (c) there are no accepted standards for measurement and compliance; (d) standards and guidelines are not enforceable because of their ambiguity; and (e) the usability of the UD resources is compromised. In effect, most UD resources are not made to support the cognitive processes of designers [3]. These shortcomings create confusion, frustration, and misinterpretation among product developers who are attempting to implement such guidelines. Thus, the UD philosophy is perceived by product developers as a special interest that slows down the time-to-market and increases costs [11,18, 23].

One approach for achieving more inclusive designs is *disability simulation*. Disability simulation is a discount method for evaluating the usability flaws in a product [12]. The tests are inexpensive and easy to implement. They are not intended to replicate the experience of living with a disability, but may help identify difficulties that those with disabilities experience when using products. Although many descriptions of simulation tests have appeared in the literature, there has been only one published validation study, involving participants who were college students [13]. Therefore, the effectiveness of this method remains unproven.

Another option is for product developers to conduct usability tests using older adults and people with disabilities [6]. While this approach yields valuable information, it can become time intensive and financially expensive. While there are many available tools and methods to evaluate usability, few have been tested on user groups with disabilities and even fewer systematically consider universal design principles [2,24,25]. Collectively, these factors are a disincentive for designers who might otherwise embrace the universal design philosophy [11,18]. There remains a clear need for efficient usability testing methods that support those who want to consider UD principles in their product designs.

This paper describes the development and preliminary testing of the Rapid Assessment of Product Usability & Universal Design (RAPUUD), a 12-item user-report tool based on the seven principles of universal design. The goal was to create a succinct, psychometrically sound tool that (a) embodies traditional usability principles, as well as UD principles, (b) lends itself to a continuum of consumer products, (c) is amenable to a diversity of user populations, including older adults and persons with disability, (d) differentiates products having different usability strengths and weaknesses, and (e) reveals product features requiring re-design.

2. Method

2.1. Study design

The validity of product usability research conducted in controlled laboratory settings is often threatened by several factors: (a) the product types and models being tested are typically identified by the investigator and may have little relevance and meaning for some participants; (b) products are often tested using contrived tasks chosen by the investigator to exemplify 'typical product usage', which may or may not reflect the task cycles that participants enact in their daily lives; (c) products are tested in settings that are unfamiliar to participants and lack the affordances present in participants' everyday environments; and (d) participants are asked to provide usability ratings based on a relatively short interval of product exposure. To circumvent these potential threats to validity, the current study employed a cross-sectional design in which participants provided one-time usability ratings on products that they had been using regularly in their everyday living environments. These ratings were gathered during nine focus groups that were conducted for a related study.

2.2. Participants

It was hoped that the RAPUUD could distinguish persons having demonstrable impairments from those having no ostensible limitations. Thus, participants were sought from four demographic groups: healthy young adults, adults with vision impairment, wheelchair users, and older adults. The young adult group was chosen as a control group. The vision im-

pairment and wheelchair user groups were chosen because they are among the largest groups having sensory limitations and physical disabilities. The older adults group was chosen because it was hoped that the RAPUUD would be responsive to the declines in physical, sensory, and cognitive function that are natural components of the aging process.

A convenience sample of each was recruited using flyers distributed to multiple locations, including: university programs in occupational therapy and human factors engineering, two offices of state vocational rehabilitation, a university-based assistive technology center, an assisted living facility, and two senior centers. Participants received a $50 honorarium in consideration of their time, which involved participation in a three-hour focus group, during which data for the current study were also collected.

The recruitment methods and experimental procedures were approved by the University at Buffalo's Social and Behavioral Sciences Institutional Review Board. Informed consent was obtained from all participants.

2.3. Instrument

The tool was developed by a four-person team of experts, all having doctoral preparation in human factors engineering. One team member is also a licensed occupational therapist, and another has a background in anthropology. Collectively, the team has extensive experience conducting research involving persons with disabilities.

2.3.1. Item development and response scaling

Item creation was based on the Principles of Universal Design [5], factors identified in a related study by Beecher and Paquet [2], usability heuristics from the human factors literature [19], and the clinical and research experiences of the co-authors. The goal was to produce a parsimonious appraisal of usability that captures product usability characteristics (e.g., physical effort, cognitive effort, visible information, safety) and elements of the task cycle (e.g., set-up, ease of use, clean-up and storage). The team followed principles of survey development suggested by DeVellis [7] and Fink [8].

For six items, the usability term was worded in a negative sense (e.g., *poses a safety risk*, *draws unwanted attention*). This was done for two reasons. First, surveys having a preponderance of positively worded items can convey the false impression that the surveyor is interested in receiving only positive feedback [7]. In addition, a mix of negatively and positively worded items can also help researchers identify respondents who provide categorically positive or negative responses.

The team considered a range of response options (e.g., visual analog, semantic differential, categorical), ultimately settling on 5-point Likert scale of endorsement (e.g., strongly agree, somewhat agree, etc.) in order to lessen respondent burden and enhance reliability. The response options also included a "not applicable" choice, since it was anticipated that not all items would be relevant for all products. The preliminary item set was iteratively refined and informally pilot tested within the development group. Ultimately, the team arrived at a 12-item version of the Rapid Assessment of Product Usability and Universal Design (RAPUUD; see Fig. 1). The content validity of RAPUUD was assessed by comparing the instrument's items to three other tools intended to aid the usability or universal design of consumer products [2,5,19].

2.4. Procedure

Each focus group included participants representing one of the target demographic groups. One moderator and one observer were present for each group. Within each group, the moderator used a four-step procedure to facilitate an ordered discussion of products used in the kitchen, home office and bathroom environments:

(1) The participants were asked to list the three products that were most challenging for them in the environment being discussed;
(2) Participants independently completed the RAPUUD tool for the product in that environment that was most challenging for them to use. The moderator and observer were available as needed to answer participant questions regarding the meaning of individual items and response options;
(3) The focus group discussion commenced, emphasizing the most frequently identified products;
(4) Participants were asked to provide comments regarding products that had not been covered during the main discussion.

The analysis discussed in this paper reflects the usability data collected during Step 2 of the above procedure. Each participant was asked to rate the usability of three products (one each for the kitchen, home office, bathroom environments) that were challenging to

1. This product is easy to set-up or prepare to use.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

2. This product is easy to use.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

3. This product is easy to clean-up and place into storage.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

4. For me, using this product poses a personal safety risk.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

5. I often need assistance to use this product

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

6. When using this product I make mistakes or errors that require me to “do over” some steps.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

7. I get the information I need to use the product efficiently.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

8. Using this product takes more time than it should.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

9. Using this product requires little physical effort.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

10. Using this product requires minimal mental effort.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

11. Using this product draws unwanted attention to me.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

12. I feel embarrassed when using this product.

☐	☐	☐	☐	☐	☐
Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree	N/a

Fig. 1. Rapid assessment of product usability and universal design (RAPUUD).

use in everyday routines at home. The self-selection of products was a deliberate choice intended to minimize the potentially biasing influence of investigator-chosen products and task cycles.

2.5. Analysis

A number of psychometric and administrative properties of the tool were evaluated, including: (a) content validity, (b) item acceptability, (c) completion time, (d) internal consistency, (e) the measurement model, including item dispersion and missing data, (f) ability to distinguish user populations, and (g) ability to distinguish product types. These reflect conventional criteria for evaluating measurement tools [1,10,21]. To facilitate statistical analysis, the ordinal scale response for each item was numerically coded on a 1-to-5 scale, with higher values denoting 'better' usability than lower values. For the positively worded items (i.e., #1–3, 7, 9, and 10), "strongly disagree" was assigned a value of 1, and "strongly agree" assigned a value of 5. For the six negatively worded items (i.e., 4–6, 8, 11, 12), "strongly disagree" was assigned a value of 5, and "strongly disagree" assigned a value of 1. The internal consistency of the tool was evaluated using the Cronbach's α statistic. Differences among user groups were evaluated using Kruskal-Wallis non-parametric analysis of variance. Post-hoc multiple comparison tests were performed to determine between-group differences once statistical significance across all groups was found. Readability statistics were calculated using a standard utility within Microsoft Word 2007 (Microsoft Corp., 2009). All other statistical analyses were completed using Minitab, v. 15.1.3.

3. Results

A total of 61 adults were recruited, representing four demographic groups: young adults ($n = 17$), older adults ($n = 22$), persons with vision impairment ($n = 12$) and wheelchair users ($n = 10$). Participant demographics (sex, age, and self-reported impairments) are summarized in Table 1. All participants rated three products, with two exceptions. One participant rated all items "not applicable" for all three products, and one participant submitted a blank rating form for the home office product. Thus, usable data were obtained on 179 products.

3.1. Content validity

An instrument is considered to have content validity to the extent that its items represent the construct being measured [20,22]. Table 2 compares the RAPUUD items to the principles articulated in three related conceptual frameworks: the Principles of Universal Design [5], Nielsen's usability heuristics [19], and factors identified by Beecher and Paquet [2]. The RAPUUD's items appear to embody the core concepts articulated in all three frameworks. In terms of clarity, the RAPUUD items are succinct (9.5 words per item), worded in the active voice (0 passive sentences), and written using plain language (4.5 characters per word; reading grade level = 6.1).

3.2. Item acceptability and completion time

Most participants completed the tool independently in 3 to 5 minutes. Participants occasionally requested clarification about the intent of individual items, most frequently item # 7 (*I get the information I need to use the product efficiently*).

3.3. Internal consistency

Collectively, the RAPUUD items achieved high internal consistency (Cronbach's $\alpha = 0.80$), which is well within the 0.70–0.90 range that is considered acceptable [22].

3.4. Measurement model

All 12 items elicited a full range of responses, with no apparent floor or ceiling effects. Figure 2 depicts the dispersion of responses for items 2, 3, 4, and 12. Several items elicited greater than 10% *Not Applicable* responses, including: *set-up*, *clean-up and storage*, *information*, *safety*, *draws unwanted attention*, and *embarrassment*. These responses are somewhat explained by the products being rated. Several product types (e.g., kitchen cabinets, desks, filing cabinets, grab bars) are typically anchored to one location and have no associated set-up or clean-up tasks. Many of these same products do not have written information or pictorial icons, in which case labeling information is also irrelevant as a usability consideration. Given that participants rated consumer products that they were regularly using at home, it is unsurprising that many participants indicated that *embarrassment* and *unwanted attention* were not applicable to their product rating.

Table 1
Participant demographics

Demographic group	Gender (Males / Females)	Self-reported impairments (n)			Mean age (Years, SD)
		Vision	Hearing	Physical	
Young adults	7 / 10	4	0	1	26.7 (3.8)
Older adults	5 / 17	10	5	8	73.4 (5.9)
Wheelchair users	5 / 5	2	0	10	48.3 (9.9)
Visually impaired	7 / 5	12	0	4	48.4 (10.4)

Table 2
Comparison of RAPUUD items to related conceptual frameworks

RAPUUD, v.1	7 Principles of universal design	Nielsen [19]	Beecher and Paquet [2]
Set-up	Size and space for approach and use.		Reach and access for use
Ease of use	Simple and intuitive use	Flexibility and efficiency of use.	Adaptability to user pace
Clean up & storage			
Safety	Tolerance for error		Secure, safe and private use Tolerance for error.
Assistance	Flexibility in use	User control & freedom Flexibility and efficiency of use	Flexibility in use
Mistakes and errors	Tolerance for error	Error prevention	Tolerance for error
Information	Perceptible information	Visibility of system status Help users recognize, diagnose, and recover from errors. Help and documentation	Perceptible information
Time		Flexibility and efficiency of use.	
Physical effort	Flexibility in use Low physical effort		Low physical effort Size for use
Mental effort	Simple and intuitive use	Match between system & real world. Consistency and standards; Recognition rather than recall;	Intuitive use Simple use
Unwanted attention	Equitable use	Esthetic and minimalist design	Equitable use Secure, safe and private use
Embarrassment	Equitable use	Esthetic and minimalist design	Equitable use

3.5. Ability to distinguish user populations

Among the four participant groups, RAPUUD scores were compared in order to assess the tool's ability to distinguish differences in usability that reflect user abilities and limitations, irrespective of product. Figure 3 summarizes the median scores on each of the RAPUUD's 12 items for the four participant groups. The shaded area of each polar graph is indicative of overall product usability. Thus, the smaller shaded areas for the wheelchair user and vision impairment groups indicate diminished usability for the assorted products rated by each of these groups.

Closer inspection reveals that individual items were also sensitive to group differences (Fig. 4). For item #5 (need for assistance) wheelchair users reported significantly lower scores then the young and older adults groups. For item #9 (physical effort) wheelchair users reported significantly lower ($p < 0.001$) scores than the other three groups. These findings would seem to reflect impairments in physical capacity that affect product usability. For item #10 (mental effort), the vision impairment group reported significantly lower scores than older adults ($p = 0.002$) and young adults ($p = 0.018$). The latter perhaps reflects the increased demands on memory, spatial awareness, and concentration that are needed to compensate for diminished visual capacity. The ratings for item #2 (ease of use) were significantly lower for the vision impairment and wheelchair user groups than for young adults ($p < 0.001$). Again, this suggests that substantial physical or sensory impairments can cause diminished product usability. The scores for item #8 (takes more time than it should) were equivalently low across the four groups, suggesting that this factor is common to sub-optimal product usability.

3.6. Ability to distinguish product types

Although participants were free to rate products of their own choosing, a number of them rated products that could be grouped into common product types (e.g.,

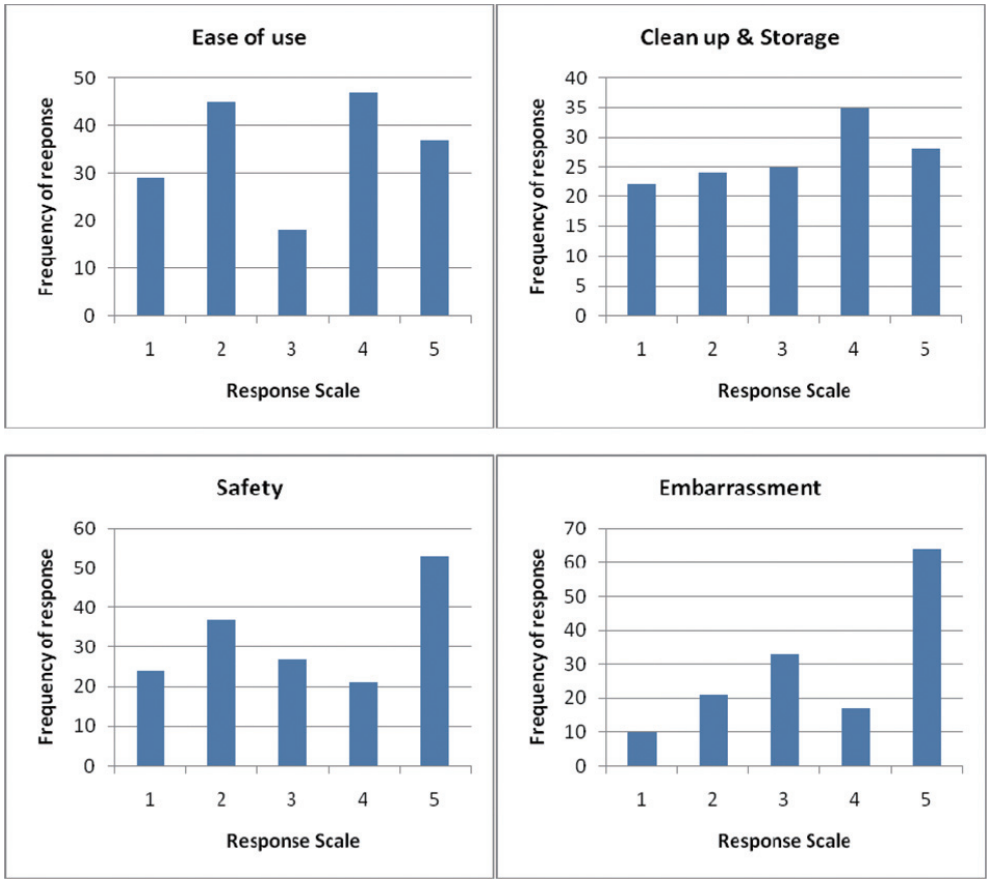


Fig. 2. Dispersion of responses for four RAPUUD items.

can openers, microwave ovens, filing cabinets, medication bottles). The four most frequently rated product types were: cabinets, showers/tubs, stoves/ovens, and computers/laptops. For each product type, the RAPUUD data were aggregated. Items having high (median score $\geqslant 4$) and low (median score $\leqslant 2$) usability ratings were identified in order to evaluate the RAPUUD's sensitivity to different product types. As summarized in Table 3, the RAPUUD scores appear to vary in accord with the demands of contrasting product types. For example, cabinets received low ratings (i.e., lower scores) for the required physical effort, safety, and amount of assistance required, which is consistent with anecdotal comments made during the focus groups. In contrast, laptop and desktop computers received high ratings in the same three areas, which is consistent with the low physical demands and risks inherent in these products. Stoves/ovens and computers/laptops received low ratings for mental effort and mistakes & errors, which is in accord with the high cognitive demands experienced by everyday users of these products. Stove/oven and shower/tub products received low ratings for safety, which is consistent with risks that are commonly acknowledged for these product types.

4. Discussion

The promising findings suggest that the RAPUUD could become a pragmatic, informative tool for researchers and designers. The data indicate that the tool was sensitive to differences in functional abilities of respondents, as well as differences in characteristics of product usability. The instrument was usable for a range of consumer products, though, as expected, not all items were appropriate for each and every product.

There were four principal limitations to this study:

1. Assessment of content validity did not include review by outside experts, which may have resulted in differences among the 12 items and response options. Nonetheless, the tool clearly

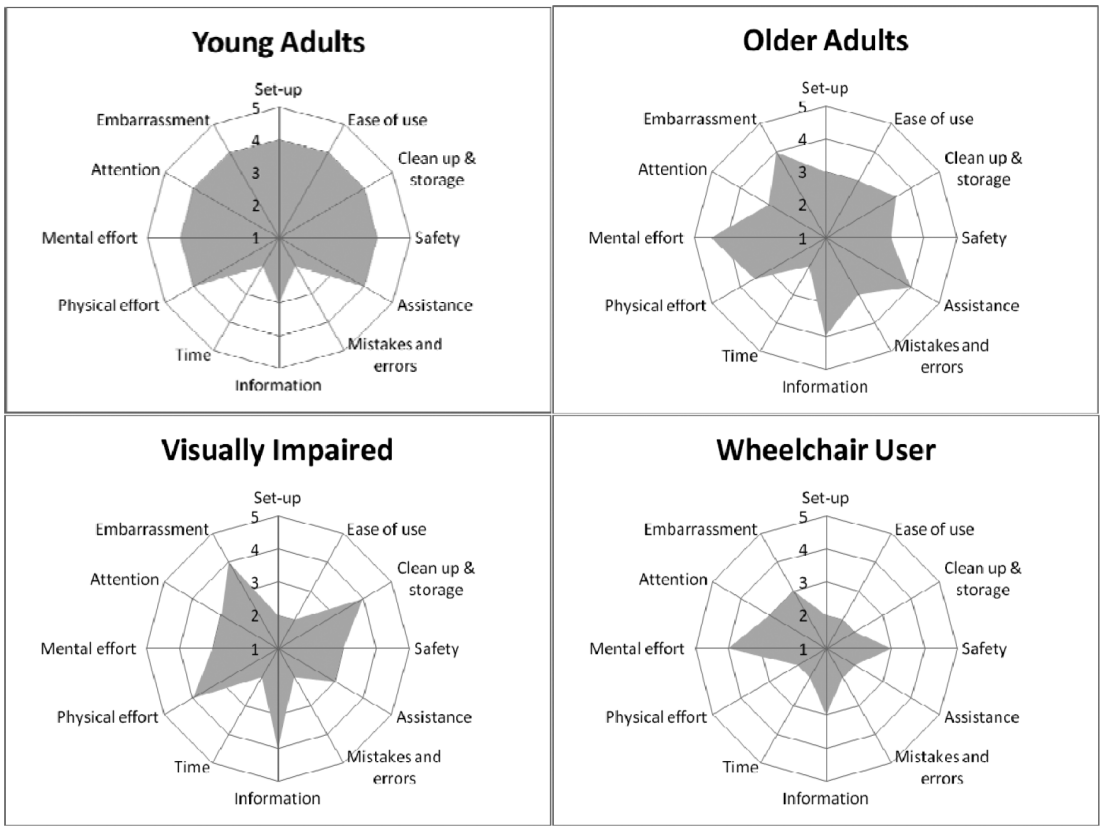


Fig. 3. Median item ratings across the four participant groups.

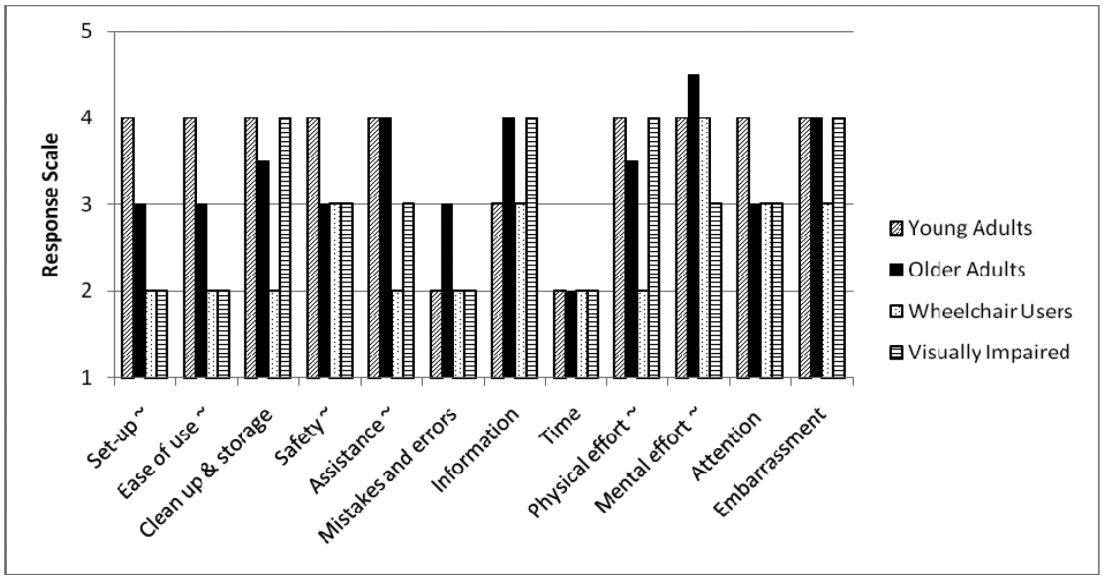


Fig. 4. Median item scores for each item and group ($\sim$ indicates significant differences between groups at $\alpha = 0.05$).

Table 3
Low- and high-rated RAPUUD items across different product types

Product type	Most problematic usability areas (RAPUUD items with median scores $\leqslant 2$)	Least problematic usability areas (RAPUUD items with median scores $\geqslant 4$)
Cabinets	Assistance	Embarrassment
($n = 22$)	Mistakes & errors	Mental Effort
	Physical effort	
	Safety	
	Set-up	
	Time	
Shower & tub	Ease of use	Assistance
($n = 19$)	Safety	Information
	Set-up	
	Time	
Stove & oven	Mental effort	Embarrassment
($n = 14$)	Mistakes & errors	Set-up
	Safety	
Computers & laptop	Mental effort	Assistance
($n = 12$)	Mistakes & errors	Embarrassment
	Set-up	Physical effort
	Time	Safety

embodies traditional usability heuristics and the principles of universal design.

2. The test-retest reliability of the RAPUUD was not evaluated. The circumstances of data collection made it infeasible to collect a second set of usability ratings from each participant. Although it is sometimes suggested that evaluation of reliability should precede evaluation of validity, the order is not essential as long as both properties are ultimately demonstrated [10].
3. The small sample size for each of the four participant groups constrains the statistical power, limiting the strength of our conclusions and muting effect sizes that might have emerged with a larger sample.

The data comparing product types and disability populations were potentially confounded by participant self-selection of products from each of the three environment categories. However, self-selection of products circumvented the potentially biasing influences of investigator-determined products and task cycles, enhancing the validity of the data in at least three ways: (a) participants rated products that had relevance in their daily lives; (b) their usability ratings reflected real-world task cycles completed in familiar environments; and (c) usability ratings reflected multiple task cycles occurring over an extended period of time. Self-selection of products also fostered a diversity of ratings, which enabled identification of potential floor and ceiling effects. Lastly, self-selection of products helped assure that the RAPUUD is, in fact, a viable tool for a variety of consumer product types.

4.1. Future research

Additional studies are needed in order to substantiate the reliability and validity of the RAPUUD, including evaluations of: (a) test-retest reliability, (b) responsiveness for additional user groups (e.g., those with hearing impairment) and populations with specific disabling conditions (e.g., persons with arthritis, stroke survivors), and (c) ability to discriminate usability concerns for novice product users. The RAPUUD must also be tested by product design teams to determine if it helps identify usability problems during product development.

Some items and response options may also benefit from modest refinement. A number of items (e.g., items 1, 4–6, and 8–10) may not truly need a "not applicable" option, since these indicators are applicable to virtually all products. In addition, some items (e.g., items 4, 6, and 12) may only require a 3-point ordinal response since safety risk, mistakes and errors, and embarrassment are factors that are either present to varying degrees or absent entirely.

5. Conclusion

This research is based on the premise that product developers already possess the technical and creative skills needed to produce accessible, usable consumer products. What they lack is insight and perspective – not for lack of caring or commitment, but for lack of exposure to the impact of functional limitations on product usage and usability. The RAPUUD tool described

here offers a practical, cost-effective technique for evaluating product usability in a manner that is consistent with the tenets of universal design and traditional usability heuristics. The RAPUUD will hopefully support product designers and developers to identify products and product features that are problematic for those with physical, sensory, and/or cognitive impairments due to aging and/or disability.

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